Before the
New York State Public Service Commission

In the Matter of

THE BROOKLYN UNION GAS COMPANY D/B/A NATIONAL GRID NY
AND
KEYSPAN GAS EAST CORPORATION D/B/A NATIONAL GRID
Cases 23-G-0225 AND Case 23-G-0226

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Prepared Direct Testimony of

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On Behalf of:

The City of New York

TABLE OF CONTENTS

I. INTRODUCTION 1

II. EXECUTIVE SUMMARY 2

A. KEDNY Summary 3

B. KEDLI Summary 7

III. REGULATORY STANDARDS 10

IV. FAIRNESS TO CUSTOMERS AND THE CLCPA 13

V. SERVICE LIFE ANALYSIS 15

A. KEDNY Accounts 19

1. Account 367 – Transmission Mains 19

2. Account 369 – Transmission Measuring and Regulating Equipment 21

3. Account 375 – Structures and Improvements 21

4. Account 376 – Distribution Mains 23

5. Account 378 – Distribution Measuring and Regulating Equipment 24

B. KEDLI Accounts 25

1. Account 367 – Transmission Mains – Steel 25

2. Account 369 – Transmission Measuring and Regulating Equipment 26

3. Account 376 – Distribution Mains 27

4. Account 378 – Distribution Measuring and Regulating Equipment 28

VI. NET SALVAGE ANALYSIS 30

VII. RESERVE VARIATION AND AMORTIZATION 34

LIST OF EXHIBITS

DJG-1	Curriculum Vitae of David J. Garrett
	KEDNY Exhibits
DJG-2	Summary Accrual and Reserve Adjustments
DJG-3	Mass Property Parameter Comparison
DJG-4	Detailed Rate and Accrual Comparison
DJG-5	Depreciation Rate Development
DJG-6	Reserve Variance Calculation
	Observed Life Tables and Iowa Curves
DJG-7	Account 367 – Transmission Mains
DJG-8	Account 369 – Transmission M&R Equipment
DJG-9	Account 375 – Structure and Improvements
DJG-10	Account 376 – Distribution Mains
DJG-11	Account 378 – Distribution M&R Equipment
DJG-12	Computed Reserve Development
	KEDLI Exhibits
DJG-13	Summary Accrual and Reserve Adjustments
DJG-14	Mass Property Parameter Comparison
DJG-15	Detailed Rate and Accrual Comparison
DJG-16	Depreciation Rate Development
DJG-17	Reserve Variance Calculation
	Observed Life Tables and Iowa Curves
DJG-18	Account 367 – Transmission Mains – Steel
DJG-19	Account 369 – Transmission M&R Equipment
DJG-20	Account 376 – Distribution Mains
DJG-21	Account 378 – Distribution M&R Equipment
DJG-22	Computed Reserve Development
DJG-23	Appendix A: The Depreciation System
DJG-24	Appendix B: Iowa Curves
DJG-25	Appendix C: Actuarial Analysis
DJG-26	Discovery Responses

I. INTRODUCTION

Q. State your name and occupation.

A. My name is David J. Garrett. I am a consultant specializing in public utility regulation. I am the managing member of Resolve Utility Consulting, PLLC.

Q. Summarize your educational background and professional experience.

A. I received a B.B.A. degree with a major in Finance, an M.B.A. degree, and a Juris Doctor degree from the University of Oklahoma. I worked in private legal practice for several years before accepting a position as assistant general counsel at the Oklahoma Corporation Commission in 2011. At the Oklahoma Commission, I worked in the Office of General Counsel in regulatory proceedings. In 2012, I began working for the Public Utility Division as a regulatory analyst providing testimony in regulatory proceedings. After leaving the Oklahoma Commission, I formed Resolve Utility Consulting, PLLC, where I have represented various consumer groups and state agencies in utility regulatory proceedings, primarily in the areas of cost of capital and depreciation. I have testified in numerous regulatory proceedings in multiple jurisdictions on the issues of cost of capital and depreciation. I am a Certified Depreciation Professional with the Society of Depreciation Professionals. I am also a Certified Rate of Return Analyst with the Society of Utility and Regulatory Financial Analysts. A more complete description of my qualifications and regulatory experience is included in my curriculum vitae, which is set forth in Exhibit ___(DJG-1).

Q. On whose behalf are you testifying in this proceeding?

A. I am testifying on behalf of the City of New York (the "City").

Q. Please describe the purpose of your testimony.

A. In my testimony, I present my analysis of the depreciation study performed by Gannett Fleming for The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, the "Companies"). I present my recommended adjustments to the Company's proposed depreciation rates and address issues raised in the Direct Testimony of Companies' witness Ned Allis, who sponsors the depreciation studies.

II. EXECUTIVE SUMMARY

Q. Please summarize the Companies' proposals regarding depreciation rates and the reserve variance.

A. The Companies are both proposing substantial increases to the annual depreciation accrual. KEDNY is proposing an annual increase of $72 million (or 50%). Exhibit__(DJG-2). KEDLI is proposing an annual increase of $32 million (or 30%). Exhibit__(DJG-13). On a percentage basis, these are among the largest proposed depreciation accrual increases I can recall from my experience in rate proceedings. The Companies' proposed increases are driven by proposed changes to service lives and net salvage rates. Because the theoretical reserve is necessarily affected by service life and net salvage rate parameters, the Companies' proposed accrual increase results in greater reserve deficiency. For

KEDNY, Mr. Allis proposes a reserve deficiency in the amount of $397.9 million. Exhibit__(DJG-2). For KEDLI, Mr. Allis proposes a reserve deficiency in the amount of $595.0 million. Exhibit__(DJG-13). The Companies propose 20-year amortizations for each of these reserve deficiencies.

A. KEDNY Summary

Q. Please provide a summary of the various adjustments you propose to the KEDNY depreciation study.

A. Based on my analyses of the KEDNY depreciations study, I would agree that some increase to the annual depreciation accrual is warranted. In fact, my proposed depreciation rates would result in an increase to the current annual depreciation accrual in the amount of $23.5 million, based on plant balances as of December 31, 2021. Exhibit__(DJG-2). My proposed adjustments would reduce KEDNY's proposed annual depreciation accrual by about $48.5 million. Exhibit__(DJG-2). My proposed adjustment addresses several issues, including: (1) extending the proposed service lives for 10 accounts; (2) decreasing the proposed negative net salvage rates for 23 accounts; and (3) not using terminal retirement dates for assets included in the Greenpoint Energy Center and Leak Prone Pipe. Consequently, these adjustments also result in a smaller reserve deficiency than the one proposed by Mr. Allis for KEDNY, resulting in no need for the amortization proposed by Mr. Allis. The collective result is a reduction in KEDNY's proposed annual depreciation accrual of approximately $60.9 million. These issues are discussed further below.

Q. Please provide a summary of your service life adjustments.

A. Regarding service life, my analysis demonstrates that the Companies' proposed service lives for several accounts are notably shorter than the service lives otherwise indicated by the Companies' historical retirement data. For KEDNY, I am proposing service life adjustments to 10 of the Company's accounts. Exhibit__(DJG-3). Each of these adjustments are based on analyzing the Company's historical retirement data under the retirement rate method, and using visual and mathematical Iowa curve-fitting techniques to estimate a reasonable service life. I provide a comparison between my proposed Iowa curves and Mr. Allis's proposed Iowa curves in my exhibits. Exhibit__(DJG-3). Out of my total proposed adjustment of $48.5 million for KEDNY, my service life adjustments account for approximately $11.9 million of that amount.

Q. Please provide a summary of your net salvage rate adjustments.

A. Mr. Allis has proposed negative net salvage rate increases for 23 KEDNY accounts. Exhibit__(DJG-3). These proposed net salvage rates significantly increase the annual depreciation accrual. I generally agree with Mr. Allis that KEDNY's historical net salvage rate data are supportive of negative net salvage rate increases for these accounts. However, I propose the Commission adopt a gradual approach regarding negative net salvage rate increases. Specifically, I propose the Commission limit KEDNY's proposed negative net salvage rate increases by 50% of the requested increase from current parameters. For example, if the currently-approved net salvage rate for an account is 0%, and the Company proposed a -20% salvage rate in this case, I would propose the Commission adopt a -10%

net salvage rate for this account in the interest of gradualism. I believe this approach is particularly prudent in light of the substantial rate increase proposed by KEDNY in this case. Out of my total proposed adjustment of $48.5 million for KEDNY, my net salvage rate adjustments account for approximately $22.0 million of that amount.

Q. Please provide a summary of your adjustments regarding the service lives of assets included in the Greenpoint Energy Center and Leak Prone Pipe.

A. In KEDNY's depreciation study, Mr. Allis applied terminal retirement dates for assets included in the Company's Greenpoint Energy Center and Leak Prone Pipe. Essentially, the Company is proposing to treat these groups of assets as "lifespan property," which is most commonly applied to electric production units and other groups of property that will retire concurrently. Otherwise, the types of assets included in the Greenpoint and Leak Prone Pipe categories would be considered "mass property," which is treated as though their utility will be an ongoing concern with no terminal retirement date. Although I am not taking issue with the Company's planned treatment for the Greenpoint Energy Center, I believe it is reasonable in this case to apply Iowa curves to each of the accounts included in this group with no specific terminal retirement date. Likewise, I propose the same treatment for the Company's leak prone pipe assets. In other words, I have used the same Iowa curves proposed by Mr. Allis for these accounts to calculate my proposed depreciation rates, but I do not apply a specific terminal retirement date under which all of the assets in this group will concurrently retire. Out of my total proposed adjustment of $48.5 million for KEDNY, my proposal to treat the Greenpoint accounts as mass property

equates to approximately $1.5 million of that amount; my proposal to treat the Leak Prone Pipe as mass property equates to approximately $13.0 million of that amount.

Q. Please provide a summary of your adjustments to KEDNY's proposed theoretical reserve and reserve deficiency allocation.

A. The theoretical reserve is necessarily impacted by current service life and net salvage proposals. Thus, unlike the book reserve, the theoretical reserve is a matter of opinion. Based on Mr. Allis's proposed service lives and net salvage rates for KEDNY's accounts, he proposes a theoretical reserve of $1.9 billion and a reserve deficiency of $398.0 million. Exhibit__(DJG-2). Because I have proposed longer service lives and lower negative net salvage rates for the accounts in dispute, I have calculated a lower theoretical reserve in the amount of $1.6 billion and a reserve deficiency of $130.4 million. Exhibit__(DJG-6). Mr. Allis proposes to amortize the amount of reserve deficiency exceeding 10% of the book reserve over 20 years, which would create an annual amortization of $12.5 million. Conceptually, I do not take issue with the proposal to amortize reserve variances exceeding 10% of the book reserve in this case, as Mr. Allis recommends. However, under my proposed depreciation parameters, the amount of the reserve deficiency does not exceed 10% of the book reserve. Exhibit__(DJG-6). Thus, I do not propose a manual reserve deficiency allocation for KEDNY in this case. If the Commission adopts depreciation rates in this case such that it results in a reserve deficiency that exceeds 10% of the book reserve, then I believe amortizing the amount in excess of 10% over a 20-year period could be reasonable depending on overall rate impacts.

B. KEDLI Summary

Q. Please provide a summary of the various adjustments you propose to the KEDLI depreciation study.

A. Based on my analyses of the KEDLI depreciations study, I would agree that some increase to the annual depreciation accrual is warranted. In fact, my proposed depreciation rates would result in an increase to the current annual depreciation accrual in the amount of $9.6 million, based on plant balances as of December 31, 2021. Exhibit__(DJG-13). My proposed adjustments would reduce KEDLI's proposed annual depreciation accrual by about $23.0 million. Exhibit__(DJG-13). My proposed adjustment addresses several issues, including: (1) extending the proposed service lives for five accounts; (2) decreasing the proposed negative net salvage rates for 30 accounts; and (3) not using terminal retirement dates for assets included in the Holtsville LNG Plant and Leak Prone Pipe. Consequently, these adjustments also result in a smaller reserve deficiency than the one proposed by Mr. Allis for KEDLI, resulting in no need for the amortization proposed by Mr. Allis. The collective result is a reduction in KEDLI's proposed annual depreciation accrual of approximately $34.4 million. These issues are discussed further below.

Q. Please provide a summary of your service life adjustments.

A. Regarding service life, my analysis demonstrates that the Companies' proposed service lives for several accounts are notably shorter than the service lives otherwise indicated by the Companies' historical retirement data. For KEDLI, I am proposing service life adjustments to five of the Company's accounts. Exhibit__(DJG-14). Each of these

adjustments are based on analyzing the Company's historical retirement data under the retirement rate method, and using visual and mathematical Iowa curve-fitting techniques to estimate a reasonable service life. I provide a comparison between my proposed Iowa curves and Mr. Allis's proposed Iowa curves in my exhibits. Exhibit__(DJG-14). Out of my total proposed adjustment of $23.0 million for KEDLI, my service life adjustments account for approximately $7.4 million of that amount.

Q. Please provide a summary of your net salvage rate adjustments.

A. Mr. Allis has proposed negative net salvage rate increases for 30 KEDLI accounts. Exhibit__(DJG-14). These proposed net salvage rates significantly increase the annual depreciation accrual. I generally agree with Mr. Allis that KEDLI's historical net salvage rate data are supportive of negative net salvage rate increases for these accounts. However, I propose the Commission adopt a gradual approach regarding negative net salvage rate increases. Specifically, I propose the Commission limit KEDLI's proposed negative net salvage rate increases by 75% of the requested increase from current parameters. For example, if the currently-approved net salvage rate for an account is 0%, and the Company proposed a -20% salvage rate in this case, I would propose the Commission adopt a -15% net salvage rate for this account in the interest of gradualism. I believe this approach is particularly prudent in light of the substantial rate increase proposed by KEDLI in this case. Out of my total proposed adjustment of $23.0 million for KEDLI, my net salvage rate adjustments account for approximately $10.8 million of that amount.

Q. Please provide a summary of your adjustments regarding the service lives of assets included in the Holtsville LNG Plant and Leak Prone Pipe.

A. In KEDLI's depreciation study, Mr. Allis applied terminal retirement dates for assets included in the Company's Holtsville LNG Plant. Essentially, the Company is proposing to treat these groups of assets as "lifespan property," which is most commonly applied to electric production units and other groups of property that will retire concurrently. Otherwise, the types of assets included in the Holtsville and Leak Prone Pipe categories would be considered "mass property," which is treated as though the utility will be a going concern with no terminal retirement date. Although I am not taking issue with the Company's planned retirement for the Holtsville LNG Plant, I believe it is reasonable in this case to apply Iowa curves to each of the accounts included in this group with no specific terminal retirement date. Likewise, I propose the same treatment for the Company's leak prone pipe assets. In other words, I have used the same Iowa curves proposed by Mr. Allis for these accounts to calculate my proposed depreciation rates, but I do not apply a specific terminal retirement date under which all of the assets in this group will concurrently retire. Out of my total proposed adjustment of $23.0 million for KEDLI, my proposal to treat the Holtsville accounts as mass property equates to approximately $0.6 million of that amount; my proposal to treat the Leak Prone Pipe assets as mass property equates to approximately $4.1 million of that amount.

Q. Please provide a summary of your adjustments to KEDLI's proposed theoretical reserve and reserve deficiency allocation.

A. The theoretical reserve is necessarily impacted by current service life and net salvage proposals. Thus, unlike the book reserve, the theoretical reserve is a matter of opinion. Based on Mr. Allis's proposed service lives and net salvage rates for KEDLI's accounts, he proposes a theoretical reserve of $1.4 billion and a reserve deficiency of $595.0 million. Exhibit__(DJG-13). Because I have proposed longer service lives and lower negative net salvage rates for the accounts in dispute, I have calculated a lower theoretical reserve in the amount of $1.2 billion and a reserve deficiency of $366.5 million. Exhibit__(DJG-17). Mr. Allis proposes to amortize the amount of reserve deficiency exceeding 10% of the book reserve over 20 years, which would create an annual amortization of $25.7 million. I also propose to amortize my calculated reserve deficiency over 20 years, which would result in an annual amortization of $14.2 million. Exhibit__(DJG-17).

III. REGULATORY STANDARDS

Q. Discuss the standard by which regulated utilities are allowed to recover depreciation expense.

A. In *Lindheimer v. Illinois Bell Telephone Co.*, the U.S. Supreme Court stated that "depreciation is the loss, not restored by current maintenance, which is due to all the factors causing the ultimate retirement of the property. These factors embrace wear and tear,

decay, inadequacy, and obsolescence."[1] The *Lindheimer* Court also recognized that the original cost of plant assets, rather than present value or some other measure, is the proper basis for calculating depreciation expense.[2] Referring to the straight-line method, the *Lindheimer* Court stated that "[a]ccording to the principle of this accounting practice, the loss is computed upon the actual cost of the property as entered upon the books, less the expected salvage, and the amount charged each year is one year's pro rata share of the total amount."[3] The original cost standard was reaffirmed by the Supreme Court in *Federal Power Commission v. Hope Natural Gas Co.* The *Hope* Court stated: "Moreover, this Court recognized in [*Lindheimer*], *supra*, the propriety of basing annual depreciation on cost. By such a procedure the utility is made whole and the integrity of its investment maintained. No more is required."[4]

The *Lindheimer* Court also held:

> [T]he company has the burden of making a convincing showing that the amounts it has charged to operating expenses for depreciation have not been excessive. That burden is not sustained by proof that its general accounting system has been correct. The calculations are mathematical, but the predictions underlying them are essentially matters of opinion.[5]

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[1] *Lindheimer v. Illinois Bell Tel. Co.*, 292 U.S. 151, 167 (1934).

[2] *Id.*

[3] *Id.* at 168.

[4] *Federal Power Commission v. Hope Natural Gas Co.*, 320 U.S. 591, 606 (1944).

[5] *Lindheimer, supra,* at 169 (emphasis added).

Thus, the Commission must ultimately determine if the Company has met its burden of proof by making a convincing showing that its proposed depreciation rates are not excessive.

Q. Discuss the definition and purpose of a depreciation system, as well as the depreciation system you employed for this project.

A. The legal standards set forth above do not mandate a specific procedure for conducting depreciation analysis. These standards, however, direct that analysts use a system for estimating depreciation rates that will result in the systematic and rational allocation of capital recovery for the utility. Over the years, analysts have developed depreciation systems designed to analyze grouped property in accordance with this standard. A depreciation system may be defined by several primary parameters: 1) a method of allocation; 2) a procedure for applying the method of allocation; 3) a technique of applying the depreciation rate; and 4) a model for analyzing the characteristics of vintage property groups.[6] In this case, I used the straight-line method, the average life procedure, the whole life technique, and the broad group model; this system would be denoted as an "SL-AL-WL-BG" system. This depreciation system conforms to the legal standards set forth above. I provide a more detailed discussion of depreciation system parameters, theories, and equations in Exhibit ___(DJG-16).

[6] *See* Wolf & W. Chester Fitch, Depreciation Systems 70, 140 (Iowa State University Press 1994).

Q. Are you using the same depreciation system that was used in the depreciation study?

A. Yes. The difference between my proposed depreciation rates and the Company's proposed rates is due to different opinions regarding service lives and reserve allocations, rather than a difference in depreciation systems.

IV. FAIRNESS TO CUSTOMERS AND THE CLCPA

Q. Please summarize the Company's position regarding the Climate Leadership and Community Protection Act as it relates to its proposed depreciation rates.

A. In his direct testimony, Mr. Allis discusses the Climate Leadership and Community Protection Act ("CLCPA") and how it informed the results of his depreciation studies and the Companies' proposal. However, Mr. Allis also states on page 6 of his pre-filed Direct Testimony that there were "no specific adjustments related to the impacts of the CLCPA."

Q. Do you agree that no specific adjustments to the Companies' depreciation rates should be made due to the CLCPA?

A. Yes. Mr. Allis makes several general conclusions in his testimony regarding the CLCPA and its potential impacts on depreciation. However, these conclusions are not supported with specific quantitative data. Even if there is a decline in gas demand as Mr. Allis suggests, there is no link provided by the Companies between such a decline in demand and the complete retirement of system assets at this time. It is also my understanding that these issues are being addressed in the New York Public Service Commission's generic long-term gas planning proceeding.

Q. Do you believe the Commission should consider the depreciation rates proposed by Mr. Allis under the units of production method?

A. No. Neither the Companies nor Mr. Allis are recommending the Units of Production depreciation rates that Mr. Allis included in his exhibits. I think for that reason alone, the Commission should not adopt these rates.

Q. Please describe your approach to estimating service life in this case in light of the Companies' testimony regarding the CLCPA.

A. Like Mr. Allis, I did not make any direct quantitative adjustment to my service life analyses to account for the impacts of the CLCPA or any other climate-related policy. However, I also did not entirely ignore this issue when using professional judgment in making my final proposed service life recommendations. Regarding the service life adjustments presented in my testimony, I primarily rely on an objective analysis of the historical data when determining the most reasonable Iowa curve for the account. In fact, the Iowa curves I select for each of the adjusted accounts below results in a better mathematical fit to the Company's observed data (the observed life table ("OLT") curve). However, none of my selected Iowa curves are the *closest* mathematically fitting curve. In fact, the Iowa curves I propose for the disputed accounts are generally *shorter* than the Iowa curve that would otherwise result in the closest mathematical fit (which would have resulted in a lower depreciation rate, all else held constant). In that regard, I am giving some consideration to Mr. Allis's general suggestions that service lives going forward may be shorter than what is otherwise strictly indicated by the historical retirement rate. Nonetheless, based on my

analysis the Company's proposed service lives for the accounts discussed below are too short given the observed data.

V. SERVICE LIFE ANALYSIS

Q. Please describe the process you used to estimate the service lives of the Company's depreciable assets.

A. To develop depreciation rates for the Company's mass property accounts, I obtained the Company's historical plant data to analyze the historical retirement patterns derived from the data. This is the same historical data used in the depreciation study. I used a set of empirically-based, standard survivor curves called "Iowa curves" to smooth and complete the observed data in order to assess the average life of each account. The depreciation study also relied upon Iowa curves in developing proposed service lives. More details about the Iowa curves are included in my exhibits. Exhibit ___(DJG-24).

Q. Please describe your approach in using Iowa curves to estimate service life.

A. The historical retirement data are used to develop an OLT for each account. The data points on the OLT can be plotted to form a curve (the "OLT curve"). The OLT curve is not a theoretical survivor curve, rather, it is a curve derived from the Company's actual historical data. An OLT curve by itself, however, is rarely a smooth curve, and it is often not a complete curve. That is, it does not end at zero percent surviving. In order to calculate average life, a complete survivor curve is needed. The Iowa curve-fitting process involves selecting the best Iowa curve to fit the OLT curve. This can be accomplished through a

combination of visual and mathematical curve-fitting techniques, as well as professional judgment. The first step of my approach to curve-fitting involves visually inspecting the OLT curve for any irregularities. For example, if the tail end of the curve is erratic and shows a sharp decline over a short period of time, it may indicate that this portion of the data is less reliable, as further discussed below. After inspecting the OLT curve, I use a mathematical curve-fitting technique which essentially involves measuring the distance between the OLT curve and the selected Iowa curve in order to get an objective, mathematical assessment of how well the curve fits. After selecting an Iowa curve, I observe the OLT curve along with the Iowa curve on the same graph to determine how well the curve fits. I may repeat this process several times for any given account to ensure that the most reasonable Iowa curve is selected. More details about the process of actuarial analysis are included in Exhibit ___(DJG-25).

Q. Do you always select the mathematically best-fitting curve?

A. Not necessarily. Mathematical fitting is an important part of the curve-fitting process because it promotes objective, unbiased results. While mathematical curve fitting is important, however, it may not always yield the optimum result; therefore, it should not necessarily be adopted without further analysis. In each of the adjusted accounts discussed below, the Iowa curve I selected would result in a closer mathematical fit to the observed data. However, for most of the accounts at issue, I did not select the closest-fitting Iowa curve from a mathematical standpoint. In each case in which I selected an Iowa curve that was not the closest fitting curve, I selected an Iowa curve that was shorter, rather than

longer, than what would otherwise be dictated by strict mathematical curve fitting. This approach highlights the balance between mathematical curve fitting and professional judgment.

Q. Please describe the additional analysis you performed for this matter.

A. After selecting an Iowa curve based on a combination of mathematical and visual curve fitting, as well as professional judgment, I applied that curve for the average life of the accounts at issue. I also considered the net salvage rate of each account in calculating my proposed depreciation rates under the whole life method. Selecting different service lives for the accounts at issue resulted in different theoretical reserve calculations for these accounts. The calculated reserve and amortization are discussed later in the testimony.

Q. Should every portion of the OLT curve be given equal weight?

A. Not necessarily. Many analysts have observed that the points comprising the tail end of the OLT curve may often have less analytical value than other portions of the curve. "Points at the end of the curve are often based on fewer exposures and may be given less weight than points based on larger samples. The weight placed on those points will depend on the size of the exposures."[7] In accordance with this standard, an analyst may decide to "truncate" the tail end of the OLT curve at a certain percent of initial exposures, such as one percent. Using this approach puts a greater emphasis on the most valuable portions of

[7] *See* Wolf & W. Chester Fitch, *supra*, at 46.

the curve. For my analysis in this case, I not only considered the entirety of the OLT curve, but also conducted further analyses that involved fitting Iowa curves to the most significant part of the OLT curve for certain accounts. To verify the accuracy of my curve selection, I also considered additional calculations and visual inspections to include truncated OLT curves based on the top 99% of the relevant data and excluding the less relevant tail-end of the OLT curve.

Q. Discuss the general differences between your service life estimates and the Company's service life estimates for the accounts to which you propose adjustments.

A. For each of the Companies' accounts in which I propose a service life adjustment, the Iowa curves I selected for these accounts result in a closer mathematical fit to the OLT curve than the Iowa curves selected by Mr. Allis. This is true for every account in dispute regardless of whether the entire OLT or truncated OLT curve is measured. This is not to say that the closest-mathematically fitting Iowa curve is always the most reasonable choice. In my opinion, for most of these accounts in dispute, the closest-fitting Iowa curve would result in an unreasonably long service life and unreasonably low depreciation rate. However, if no convincing evidence is presented outside of the historical retirement data, I believe service life estimates should be primarily based on objective statistical analyses, while using judgment to ensure that the service life estimate is not unreasonably long short based on the other information presented in the depreciation study and consideration of service lives typically observed in the industry for the same accounts. The analysis of each selected account is presented below.

A. KEDNY Accounts

1. Account 367 – Transmission Mains

Q. Describe your service life estimate for this account and compare it with the Company's proposal.

A. The OLT curve for this account and other accounts discussed in this section is constructed using the Company's historical property data. The graph shown in Exhibit ___(DJG-7) for this account shows the two different Iowa curves selected by the Company and me to best represent the average remaining life for the assets in this account. For this account, Mr. Allis selected the R2-75 Iowa curve, and I selected the R2.5-85 Iowa curve. Both Iowa curves are displayed along with the OLT curve for this account in a graph included in Exhibit ___(DJG-7).[8] The vertical dotted line in this graph represents the truncation point based on the 1% truncation benchmark discussed above. The data points on the OLT curve occurring to the right of this line are statistically less relevant than the other data points on the OLT curve due to the relatively low amount of dollars exposed to retirement associated with data points at the tail-end of the OLT curve. As shown in the graph, the relative lack of retirement experience does not provide a confident indication of the retirement dispersion pattern (*i.e.*, Iowa curve shape), however, much of the data on this OLT curve is nonetheless statistically relevant. In addition, more than 90% of the dollars older than

[8] The depreciation study consolidates the following accounts for service life analysis and applies the same Iowa curve to each account: 367.06 – Plastic Mains, 367.09 – Steel Mains (Post 1991), and 367.08 – Steel Mains (Pre 1992). I also apply the same Iowa curve for each of these accounts.

100 years are still surviving in this account. The Iowa curve I selected for this account strikes a fair balance between the information presented in the depreciation study and testimony, and the much longer service life indicated by the statistical data. Although it is not pertinent for this particular account, we can use mathematical curve fitting techniques to further assess the results.

Q. Does your selected curve provide a better mathematical fit to the truncated OLT curve than the Company's selected Iowa curve for this account?

A. Yes. Selected Iowa curves based on visual curve fitting techniques can be confirmed and bolstered by checking them mathematically. The better mathematically-fitted curve is the one that minimizes the distance between the OLT curve and the Iowa curve, thus providing the closest fit. The "distance" between the curves is calculated using the "sum-of-squared differences" ("SSD") technique. The curve with the lower SSD represents the better mathematical fit. For this account, the SSD between the Company's Iowa curve and the truncated OLT curve is 1.0560, and the SSD between the R2.5-85 curve I selected and the OLT curve is 0.3689, which means it results in the closer fit. These calculations are shown in Exhibit ___(DJG-7).

Q. Describe the impact to KEDY's proposed depreciation accrual for this account if your recommended service life is adopted.

A. Adopting my proposed adjustments for this account would result in an adjustment reducing KEDNY's proposed depreciation accrual by $2,615,179.

2. Account 369 – Transmission Measuring and Regulating Equipment

Q. Describe your service life estimate for this account and compare it with the Company's estimate.

A. For this account, Mr. Allis selected the R3-50 curve, and I selected the R3-58 curve. A graph included in Exhibit ___(DJG-8) shows these two curves along with the OLT curve. Although the OLT curve derived from the historical retirement data for this account is less than ideal given the amount of retirement experience, the relevant data nonetheless indicate a longer average service life than the one proposed by Mr. Allis.

Q. Does your selected Iowa curve provide a better mathematical fit to the truncated OLT curve for this account?

A. Yes. The SSD between the Company's Iowa curve and the truncated OLT curve is 0.0675, and the SSD between the R3-58 curve I selected and the OLT curve is 0.0221, which means it results in the closer fit. These calculations are shown in Exhibit ___(DJG-8).

Q. Describe the impact to KEDY's proposed depreciation accrual for this account if your recommended service life is adopted.

A. Adopting my proposed adjustments for this account would result in an adjustment reducing KEDNY's proposed depreciation accrual by $350,511.

3. Account 375 – Structures and Improvements

Q. Describe your service life estimate for this account and compare it with the Company's estimate.

A. For this account, Mr. Allis selected the R3-60 curve, and I selected the R3-75 curve. A graph included in Exhibit ___(DJG-9) shows these two curves along with the OLT curve. The OLT curve for Account 375 is well suited for conventional Iowa curve fitting techniques. That is, it has a sufficient amount of retirement history, is relatively smooth, and most of the curve is statistically relevant. As shown in the graph in Exhibit___(DJG-9), the Iowa curve proposed by Mr. Allis is significantly shorter than the average life otherwise indicated by the Company's historical retirement data as presented in the OLT curve. Even the R2-75 Iowa curve I propose is shorter than the indicated service life at this time, however, the longer life and flatter trajectory of the R2-75 results in a more reasonable fit to the historical data at this time. Effectively, both Iowa curves "anticipate" that the retirement rate going forward in this account may be greater than the past retirement rate, causing the OLT curve to decline. However, it is not reasonable to make a projection of remaining life that is so divergent from the historical retirement pattern as Mr. Allis's selected Iowa curve does.

Q. Does your selected Iowa curve provide a better mathematical fit to the truncated OLT curve for this account?

A. Yes. The SSD between the Company's Iowa curve and the truncated OLT curve is 9.9549, and the SSD between the R2-75 curve I selected and the OLT curve is 1.7328, which means it results in the closer fit. These calculations are shown in Exhibit ___(DJG-9).

Q. Describe the impact to KEDY's proposed depreciation accrual for this account if your recommended service life is adopted.

A. Adopting my proposed adjustments for this account would result in an adjustment reducing KEDNY's proposed depreciation accrual by $595,832.

4. Account 376 – Distribution Mains

Q. Describe your service life estimate for this account and compare it with the Company's estimate.

A. For this account, Mr. Allis selected the R2-75 curve, and I selected the R1.5-84 curve. A graph included in Exhibit ___(DJG-10) shows these two curves along with the OLT curve.[9] The OLT curve for Account 376 is well suited for conventional Iowa curve fitting techniques. That is, it has a sufficient amount of retirement history, is relatively smooth and complete (falling to nearly 30% surviving). As shown in the graph in Exhibit___(DJG-10), the Iowa curve proposed by Mr. Allis provides a relatively close fit to the truncated OLT curve, and then diverges below the from the OLT curve after that point. Unlike the previous account, I do not believe Mr. Allis's proposed Iowa curve is outside the range of reasonableness. However, both the truncated and entire OLT curve indicate a longer average serve life with a flatter trajectory that is more accurately reflected in the R1.5-84 Iowa curve I selected. We can confirm the results mathematically.

[9] The depreciation study consolidates the following accounts for service life analysis and applies the same Iowa curve to each account: 376.01 – Steel Mains (Post 1991), 376.07 – Plastic Mains (Pre 1992), 376.09 – Steel Mains (Pre 1992), and 376.18 – Plastic Mains (Post 1991). I also apply the same Iowa curve for each of these accounts.

Q. Does your selected Iowa curve provide a better mathematical fit to the truncated OLT curve for this account?

A. Yes. The SSD between the Company's Iowa curve and the truncated OLT curve is 0.0322, and the SSD between the R1.5-84 curve I selected and the OLT curve is 0.0212, which means it results in the closer fit. These calculations are shown in Exhibit ___(DJG-10).

Q. Describe the impact to KEDY's proposed depreciation accrual for this account if your recommended service life is adopted.

A. Adopting my proposed adjustments for this account would result in an adjustment reducing KEDNY's proposed depreciation accrual by $26,123,800.

5. Account 378 – Distribution Measuring and Regulating Equipment

Q. Describe your service life estimate for this account and compare it with the Company's estimate.

A. For this account, Mr. Allis selected the R2.5-50 curve, and I selected the R2-58 curve. A graph included in Exhibit ___(DJG-11) shows these two curves along with the OLT curve. The OLT curve for Account 376 is fairly well suited for conventional Iowa curve fitting techniques. Even the truncated OLT curve drops to 60% surviving, giving a fairly good indication of the retirement pattern and average life. The Iowa curve selected by Mr. Allis is clearly shorter than the retirement pattern indicated in the OLT curve. We can use mathematical curve fitting to further assess the results.

Q. Does your selected Iowa curve provide a better mathematical fit to the truncated OLT curve for this account?

A. Yes. The SSD between the Company's Iowa curve and the truncated OLT curve is 1.3542, and the SSD between the R2-58 curve I selected and the OLT curve is 0.1739, which means it results in the closer fit. These calculations are shown in Exhibit ___(DJG-11).

Q. Describe the impact to KEDY's proposed depreciation accrual for this account if your recommended service life is adopted.

A. Adopting my proposed adjustments for this account would result in an adjustment reducing KEDNY's proposed depreciation accrual by $810,992.

B. KEDLI Accounts

1. Account 367 – Transmission Mains – Steel

Q. Describe your service life estimate for this account and compare it with the Company's estimate.

A. For this account, Mr. Allis selected the R2.5-75 curve, and I selected the R2.5-85 curve. A graph included in Exhibit ___(DJG-18) shows these two curves along with the OLT curve. As shown in the graph, the relative lack of retirement experience does not provide a confident indication of the retirement dispersion pattern (*i.e.*, Iowa curve shape), however, much of the data on this OLT curve is nonetheless statistically relevant. In addition, more than 90% of the dollars older than 100 years are still surviving in this account. The Iowa curve I selected for this account strikes a fair balance between the information presented in the depreciation study and testimony, and the much longer service life indicated by the

statistical data. Although it is not pertinent for this particular account, we can use mathematical curve fitting techniques to further assess the results.

Q. Does your selected Iowa curve provide a better mathematical fit to the truncated OLT curve for this account?

A. Yes. The SSD between the Company's Iowa curve and the truncated OLT curve is 1.0560, and the SSD between the R2.5-85 curve I selected and the OLT curve is 0.3689, which means it results in the closer fit. These calculations are shown in Exhibit ___(DJG-18).

Q. Describe the impact to KEDLI's proposed depreciation accrual for this account if your recommended service life is adopted.

A. Adopting my proposed adjustments for this account would result in an adjustment reducing KEDLI's proposed depreciation accrual by $1,792,361.

2. <u>Account 369 – Transmission Measuring and Regulating Equipment</u>

Q. Describe your service life estimate for this account and compare it with the Company's estimate.

A. For this account, Mr. Allis selected the R3-45 curve, and I selected the R2.5-54 curve. A graph included in Exhibit ___(DJG-19) shows these two curves along with the OLT curve. The OLT curve for Account 369 is fairly well suited for conventional Iowa curve fitting techniques. Even the truncated OLT curve drops to nearly 60% surviving, giving a fairly good indication of the retirement pattern and average life. The Iowa curve selected by Mr. Allis is clearly shorter than the retirement pattern indicated in the OLT curve. Even when the truncated OLT curve is considered, the Iowa curve selected by Mr. Allis does not

provide an accurate fit from about ages 30-55, which is a significant, relevant portion of the OLT curve. As a result, I believe his proposed depreciation rate for this account is unreasonably high. We can use mathematical curve fitting to further assess the results.

Q. Does your selected Iowa curve provide a better mathematical fit to the truncated OLT curve for this account?

A. Yes. The SSD between the Company's Iowa curve and the truncated OLT curve is 1.2425, and the SSD between the R2.5-54 curve I selected and the OLT curve is 0.0767, which means it results in the closer fit. These calculations are shown in Exhibit ___(DJG-19).

Q. Describe the impact to KEDLI's proposed depreciation accrual for this account if your recommended service life is adopted.

A. Adopting my proposed adjustments for this account would result in an adjustment reducing KEDLI's proposed depreciation accrual by $681,005.

3. Account 376 – Distribution Mains

Q. Describe your service life estimate for this account and compare it with the Company's estimate.

A. For this account, Mr. Allis selected the R2.5-75 curve, and I selected the R2-82 curve. A graph included in Exhibit ___(DJG-20) shows these two curves along with the OLT curve.[10] The OLT curve for Account 376 is well suited for conventional Iowa curve fitting

[10] The depreciation study consolidates the following accounts for service life analysis and applies the same Iowa curve to each account: 376.08 – Steel Mains and 376.18 – Plastic Mains. I also apply the same Iowa curve for each of these accounts.

techniques. That is, it has a sufficient amount of retirement history, is relatively smooth and complete (falling to nearly 5% surviving). As shown in the graph in Exhibit___(DJG-20), the Iowa curve proposed by Mr. Allis provides a relatively close fit to the truncated OLT curve, and then diverges below the from the OLT curve after that point. Unlike the previous account, I do not believe Mr. Allis's proposed Iowa curve is outside the range of reasonableness. However, both the truncated and entire OLT curve indicate a longer average serve life with a flatter trajectory that is more accurately reflected in the R2-82 Iowa curve I selected.

Q. Does your selected Iowa curve provide a better mathematical fit to the truncated OLT curve for this account?

A. Yes. The SSD between the Company's Iowa curve and the truncated OLT curve is 0.0107, and the SSD between the R2-82 curve I selected and the OLT curve is 0.0027, which means it results in the closer fit. These calculations are shown in Exhibit ___(DJG-20).

Q. Describe the impact to KEDLI's proposed depreciation accrual for this account if your recommended service life is adopted.

A. Adopting my proposed adjustments for this account would result in an adjustment reducing KEDLI's proposed depreciation accrual by $11,510,584.

4. <u>Account 378 – Distribution Measuring and Regulating Equipment</u>

Q. Describe your service life estimate for this account and compare it with the Company's estimate.

A. For this account, Mr. Allis selected the R3-40 curve, and I selected the R2.5-55 curve. A graph included in Exhibit ___(DJG-21) shows these two curves along with the OLT curve. The OLT curve for Account 378 is well suited for conventional Iowa curve fitting techniques. That is, it has a sufficient amount of retirement history, is relatively smooth, and has sufficient retirement experience. As shown in the graph in Exhibit___(DJG-21), the Iowa curve proposed by Mr. Allis is significantly shorter than the average life otherwise indicated by the Company's historical retirement data as presented in the OLT curve. Even the R2.5-55 Iowa curve I propose is also shorter than the indicated service life at this time, however, the longer life and flatter trajectory of the R2.5-55 results in a more reasonable fit to the historical data at this time. Effectively, both Iowa curve "anticipate" that the retirement rate going forward in this account may be greater than the past retirement rate, causing the OLT curve to decline. However, it is not reasonable to make a projection of remaining life that is so divergent from the historical retirement pattern as Mr. Allis's selected Iowa curve does.

Q. Does your selected Iowa curve provide a better mathematical fit to the truncated OLT curve for this account?

A. Yes. The SSD between the Company's Iowa curve and the truncated OLT curve is 3.7080, and the SSD between the R2.5-55 curve I selected and the OLT curve is 0.1562, which means it results in the closer fit. These calculations are shown in Exhibit ___(DJG-21).

Q. Describe the impact to KEDLI's proposed depreciation accrual for this account if your recommended service life is adopted.

A. Adopting my proposed adjustments for this account would result in an adjustment reducing KEDLI's proposed depreciation accrual by $909,897.

VI. NET SALVAGE ANALYSIS

Q. Please describe the concept of net salvage.

A. If an asset has any value left when it is retired from service, a utility might decide to sell the asset. The proceeds from this transaction are called "gross salvage." The corresponding expense associated with the removal of the asset from service is called the "cost of removal." The term "net salvage" equates to gross salvage less the cost of removal. Often, the net salvage for utility assets is a negative number (or percentage) because the cost of removing the assets from service exceeds any proceeds received from selling the assets. When a negative net salvage rate is applied to an account to calculate the depreciation rate, it results in increasing the total depreciable base to be recovered over a particular period of time and increases the depreciation rate. Therefore, a greater negative net salvage rate equates to a higher depreciation rate and expense, all else held constant.

Q. Please describe the Companies' proposal regarding its net salvage rates for mass property accounts.

A. Mr. Allis has proposed negative net salvage rate increases for 23 KEDNY accounts. Exhibit__(DJG-3). Mr. Allis also proposed negative net salvage rate increases for 30 KEDLI accounts. Exhibit__(DJG-14). These proposed net salvage rates significantly increase the annual depreciation accrual.

Q. Do you agree with the Companies' proposed net salvage rate increases?

A. I agree that the historical net salvage data provided in the depreciation study are generally supportive of the negative net salvage rate increases proposed by Mr. Allis. However, I believe that these increases contribute to an overall depreciation accrual increase that is unreasonable and overly burdensome to customers. Combined, the Companies are proposing an increase of more than $100 million (more than 40%) to the annual depreciation accrual. Even if our proposed reduced accruals are adopted, this increase to the net salvage rate remains concerning due to the ongoing increased financial burden it poses to customers.

Q. What do you propose regarding net salvage?

A. In the interest of fairness to customers, I propose the Commission consider gradualism when adopting the proposed negative net salvage increases in this case. Specifically, I recommend the Commission limit KEDNY's proposed negative net salvage rate increases by 50% of the requested annual increase from current parameters. In addition, I recommend the Commission limit KEDLI's proposed negative net salvage rate increases by 75% of the requested annual increase from current parameters.

Q. Do you think that reducing the accruals could cause a cash flow issue for the utilities?

A. No, I do not think it would have a negative impact on cash flow. My proposal would still result in a net increase in cash flow for both utilities, all else held constant.

Q. Has there been a trend in increasing negative net salvage in the utility industry?

A. Yes. Negative net salvage rates occur when the cost of removal exceeds the gross salvage of an asset when it is removed from service. Net salvage rates are calculated by considering gross salvage and removal costs as a percent of the original cost of the assets retired. In other words, salvage and removal costs are based on current dollars, while retirements are based on historical dollars. Increasing labor costs associated with asset removal combined with the fact that original costs remain the same have contributed to increasing negative net salvage over time.

Q. Have other utility commissions expressed concern over increasing negative net salvage rates?

A. Yes. In the Pacific Gas and Electric Company's ("PG&E") 2014 rate case, the California commission stated: "We remain concerned with the growing cost burden associated with increasing cost trends for negative net salvage."[11] The California commission also expressed an interest in the ratemaking concept of gradualism:

[11] Decision Authorizing Pacific Gas and Electric Company's General Rate Case Revenue Requirement for 2014-2016, D.14-08-032, p. 597

> In evaluating whether a proposed increase reflects gradualism, however, we believe the more appropriate measure is how the change affects customers' retail rates. The fact that PG&E previously proposed higher removal costs than adopted has no bearing on how a proposed change would impact current ratepayers. Accordingly, we apply the principle of gradualism based on how a proposed change in estimate compares to adopted costs reflected in current rates, irrespective of what PG&E may have forecasted in an earlier depreciation study.[12]

In PG&E's 2014 rate case, the Office of Ratepayer Advocates proposed a 25% cap on increased net salvage rates to mitigate sudden increases in net salvage and instead provide for more gradual levels of increases.[13] The Commission ultimately found: "As a general approach, we adopt no more than 25% of PG&E's estimated increases in the accrual provision for removal costs. This limitation tempers the impacts on current ratepayers. . . ."[14]

Q. Are you proposing a gradual approach for the accounts to which the Companies propose decreases to negative net salvage rates?

A. No. My proposed net salvage adjustments under the gradualism principle apply only to the accounts to which the Companies' have proposed negative net salvage rate increase, not decreases. This is because the gradualism principle I am proposing is designed to mitigate the immediate financial impact to current customers. Even if all of my recommended adjustments to the Companies' proposed depreciation rates were adopted, it would still result in overall accrual increases for both Companies. Under this scenario, it

[12] *Id*. at 598.
[13] *Id*. at 592-93.
[14] *Id*. at 602.

would not make as much sense in terms of fairness to apply the gradualism protection in the interest of shareholders.

VII. RESERVE VARIATION AND AMORTIZATION

Q. Please describe the issue concerning the reserve variation.

A. At any given point in time, there will likely be a discrepancy between the recorded book reserve and the theoretical reserve. The theoretical reserve is a balance that would theoretically be in the accumulated depreciation account using the current proposed depreciation parameters (service life and salvage). When depreciation rates are calculated under the remaining life technique, any imbalance or variation between the book reserve and theoretical reserve is essentially allocated automatically over the remaining life of plant. In this case, since proposed depreciation rates are developed under the whole life technique, it is necessary to address the reserve variation as a separate issue.

Q. Please describe the Companies' calculated reserve variation and proposed amortization.

A. For KEDNY, Mr. Allis proposes a theoretical reserve of $1.9 billion and a reserve deficiency of $398.0 million for KEDNY. Exhibit___(DJG-2). Mr. Allis proposes to amortize the amount of reserve deficiency exceeding 10% of the book reserve over 20 years, which would create an annual amortization of $12.5 million. For KEDLI, he proposes a theoretical reserve of $1.4 billion and a reserve deficiency of $595.0 million. Exhibit__(DJG-13). Mr. Allis proposes to amortize the amount of reserve deficiency

exceeding 10% of the book reserve over 20 years, which would create an annual amortization of $25.7 million.

Q. Please describe your proposed adjustments to the Companies' calculated reserve variation and proposed amortization.

A. Because I propose longer service lives and lower negative net salvage rates or both Companies, my theoretical reserve calculation will necessarily be lower for both Companies. For KEDNY, I calculate a theoretical reserve in the amount of $1.6 billion and a reserve deficiency of $130.4 million. Conceptually, I do not take issue with the proposal to amortize reserve variances exceeding 10% of the book reserve in this case, as Mr. Allis recommends. However, under my proposed depreciation parameters, the amount of the reserve deficiency does not exceed 10% of the book reserve. Exhibit__(DJG-6). Thus, I do not propose a manual reserve deficiency allocation for KEDNY in this case. For KEDLI, I have calculated a lower theoretical reserve in the amount of $1.2 billion and a reserve deficiency of $366.5 million. Exhibit__(DJG-17). I also propose to amortize my calculated reserve deficiency over 20 years, which would result in an annual amortization of $14.2 million. Exhibit__(DJG-17).

Q. Does this conclude your testimony?

A. Yes.